MarketBite: Dynamic Pricing for Campus Dining

Pricing Model and Methodology

I. Prototype Pricing Process

Our prototype prices surplus food using available stock and time before its selling deadline. Cafe Desk controls the shared simulation for the Food Display, Exchange Display, and Mobile Display. All three use the same confirmed prices within the owner's limits.

II. Inputs and Mathematical Notation

A. Price and Available Stock

The regular price (P) is the undiscounted dollar price per item. The starting quantity (q_0) is the number received in a batch. It stays fixed as items sell; each delivery starts a new batch with its own starting count.

The available quantity (q) equals physical stock minus reserved items, protected items, and the inventory buffer. Committed or set-aside food is excluded. For a positive starting quantity, the stock share (S) is:

$$S = \min\left\{1, \frac{q}{q_0}\right\} \tag{1}$$

Thus, 10 available items from a batch of 20 give $S = 0.50$. The cap at 1 prevents the share from exceeding 100%; empty batches are withdrawn.

B. Time and Monetary Rounding

The elapsed time (t) runs from batch arrival to the review. The selling window (W) runs from arrival to the deadline, in the same time units. For a positive window, the elapsed-time share (T) is:

$$T = \min\left\{1, \max\left\{0, \frac{t}{W}\right\}\right\} \tag{2}$$

Halfway through the window means $T = 0.50$. The minimum and maximum functions bound the share between 0 and 1. Time advances between reviews while the confirmed price remains unchanged.

The upward cent-rounding operator (U) rounds a nonnegative dollar amount (x) to the next cent, leaving exact cent values unchanged. The ceiling brackets denote rounding upward to an integer:

$$\mathrm{U}(x) = \frac{\lceil 100x \rceil}{100} \tag{3}$$

III. Calculate the Target Price

The calculation finds a discount from stock and time, then enforces the owner's minimum price. All percentages enter the equations as decimals: for example, 45% is written as 0.45.

A. Discount and Target Before the Floor

The maximum discount (D) is 0.45 by default and can range from 0 to 0.45. The calculated discount (d) combines this cap with stock and time as follows:

$$d = D\,S\,T \tag{4}$$

The target before applying a floor (R) is the regular price after that discount, rounded upward to cents:

$$R = \mathrm{U}(P(1 - d)) \tag{5}$$

For fixed stock, elapsed time increases the discount. For fixed time, falling stock can raise the target. The 45% cap is a limit, not a promised closing discount.

B. Cost Minimum and Price Floor

The selling cost (c), fixed order-fee allowance (o), and required contribution (m) are dollar amounts used for each item's floor. Contribution is the amount the cafe requires to retain after these configured costs and fees. The percentage fee rate (f) is a nonnegative decimal below 1. The cost minimum (E) is:

$$E = \mathrm{U}\left(\frac{c + o + m}{1 - f}\right) \tag{6}$$

The owner's minimum (M) can impose a higher limit. The effective price floor (F) is the largest of one cent, that minimum, the discount-cap minimum, and the cost minimum required for each sale:

$$F = \max\{0.01, M, \mathrm{U}(P(1 - D)), E\} \tag{7}$$

With the default cap, the discount-cap minimum is 55% of the regular price. If F exceeds P, the offer pauses. Otherwise, the bounded target (P^*) is:

$$P^* = \max\{F, R\} \tag{8}$$

The target stays between the floor and regular price. The review then applies the adjustment limit before confirming the displayed price, as described on page 2.

IV. Review and Confirm the Price

Cafe Desk review options are 5, 30, or 60 minutes. The default is 30 simulated minutes. Purchases update stock immediately; automatic repricing waits for a scheduled review and uses the latest stock and time.

A. Limit Each Adjustment

The step fraction (a) ranges from 0.01 to 0.05, with 0.05 as the default. The dollar step (Δ) is based on the regular price, so it does not shrink as discounts accumulate:

$$\Delta = \mathrm{U}(aP) \tag{9}$$

For review index (k), the current confirmed price (p_k) moves toward the bounded target. The provisional price (z) is limited to one step in either direction:

$$z = \min\left\{p_k + \Delta, \max\left\{p_k - \Delta, P^*\right\}\right\} \tag{10}$$

The next confirmed price (p_{k+1}) also respects the floor and regular-price ceiling:

$$p_{k+1} = \max\left\{F, \min\left\{P, z\right\}\right\} \tag{11}$$

The price may fall, rise, or hold. A new floor can require an increase beyond the usual step, taking priority when these limits conflict.

B. Share the Confirmed Price

The simulation must confirm a price update; failure leaves the previous price in place. When several eligible batches offer an item, displays show the lowest confirmed batch price. Purchases record the price paid at that time.

V. Worked Price Adjustment

Assume P = \$10, q_0 = 20, q = 10, and T = 0.50. The current price is \$10, the floor is \$6, and default discount and step settings apply. Equation (1) gives S = 0.50, so:

$$d = 0.45 \times 0.50 \times 0.50 = 0.1125$$

$$P^* = \max\{6.00, \mathrm{U}(8.875)\} = 8.88$$

The unrounded target discount is 11.25%. Rounding \$8.875 upward gives \$8.88, or 11.2% off, above the \$6 floor. The step and confirmed price are:

$$\Delta = \mathrm{U}(0.05 \times 10.00) = 0.50$$

$$p_{k+1} = 10.00 - 0.50 = 9.50$$

All three displays show \$9.50: an actual discount of 5%. Later reviews recalculate the target from updated inputs; future discounts are not predetermined.

VI. Costs and Price Increases

The examples below use default settings to show how the target, floor, and step interact.

A. Calculate a Cost Floor

Suppose c = \$6, o = \$0.20, m = \$0.50, and the percentage fee is 3%, so f = 0.03. Equation (6) gives:

$$E = \mathrm{U}\left(\frac{6.00 + 0.20 + 0.50}{1 - 0.03}\right) = 6.91$$

For P = \$10, the default cap sets a \$5.50 minimum. If the owner requires \$7, equation (7) selects:

$$F = \max\{0.01, 7.00, 5.50, 6.91\} = 7.00$$

A lower target rises to \$7 before the review step. This floor covers the entered assumptions; it does not establish the profitability of the whole cafe.

B. Understand Upward Adjustments

With P = \$10, 4 of 20 items available, and 75% of the window elapsed, the unrounded target discount is 6.75%. The target before the floor rounds up from \$9.325 to \$9.33.

From \$8.80 with a lower floor, the \$0.50 step allows \$9.30. Even near closing, scarcity can raise prices: one of 20 items left with six minutes of a 120-minute window remaining gives a \$9.79 target. A closing clearance discount is not guaranteed.

A new \$9.60 floor raises a \$9.00 current price to at least \$9.60 under equation (11), overriding the normal \$0.50 step. The regular price remains the upper limit.

VII. Availability and Interpretation

Offers pause if stock is unavailable, counts are more than 30 minutes old, sale approval or checkout support is absent, or the floor exceeds the regular price. Missing or unreviewed costs require review. Batches ineligible for surplus pricing keep their confirmed price.

New orders stop five minutes before the selling deadline. This buffer is separate from the review interval. Cafe Desk controls inventory and pricing settings.

More frequent reviews allow more steps toward the target during the same period; they do not change the formula, step size, or 30-minute inventory freshness limit.

Identical inputs produce identical results. Purchases and confirmations are simulated. The prototype's effects on student savings, edible food waste, and cafe profitability must be measured in a live pilot.